Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, Illia Polosukhin

Google Brain; Google Research; University of Toronto

The Transformer replaces recurrence and convolution with attention, improving translation quality while enabling more parallel training and transferring successfully to constituency parsing.

Introduction

- Recurrent encoder–decoder models process sequence positions sequentially, limiting parallelization within training examples.
- Convolutional models enable parallel computation, but connecting distant positions requires multiple layers.
- Self-attention directly connects sequence positions, shortening paths for learning long-range dependencies.

Objectives

- Develop a sequence transduction architecture without recurrent or convolutional layers.
- Evaluate translation quality and training cost against published recurrent and convolutional models.
- Test generalization to English constituency parsing with limited and large training datasets.

Methods

- Train on WMT 2014 English–German: approximately 4.5 million sentence pairs; English–French: 36 million sentence pairs, using subword tokenization.
- Stack six encoder and six decoder layers; combine multi-head attention, position-wise feed-forward networks, residual connections, and layer normalization.
- Compute attention as softmax($QK^T/\sqrt{d_k}$)V, using query, key, and value matrices; the base model has eight attention heads.
- Add sinusoidal positional encodings; mask future decoder positions and attend to encoder outputs, preserving autoregressive generation.
- Train with Adam, learning-rate warmup, dropout, and label smoothing on eight NVIDIA P100 GPUs; evaluate translation using beam search and checkpoint averaging.

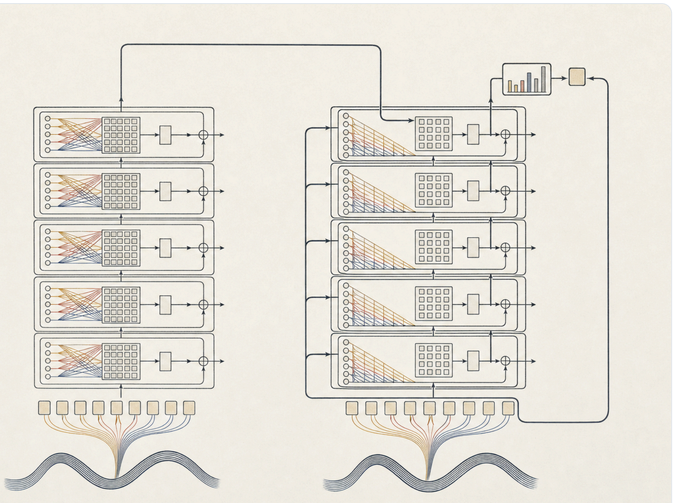


Figure 1. Transformer encoder–decoder schematic: positional information enters both stacks; parallel attention heads connect sequence positions, while masked decoder attention preserves autoregressive prediction.

Results

28.4
English–German test BLEU

41.8
English–French BLEU, Abstract and Table 2

3.5 days
Big-model training on eight P100 GPUs

- English–German: Transformer big achieves 28.4 BLEU versus 26.36 for the previously reported ConvS2S ensemble; base achieves 27.3.
- English–French: Abstract and Table 2 report 41.8 BLEU versus 40.56 for the prior single-model MoE; Section 6.1 instead reports 41.0.
- Base-model training takes 12 hours; big-model training takes 3.5 days, both on eight NVIDIA P100 GPUs.
- English–German development ablation: eight attention heads achieve 25.8 BLEU versus 24.9 with one head; learned positional embeddings achieve 25.7.
- English constituency parsing: four-layer Transformer achieves 91.3 F1 with Wall Street Journal data only and 92.7 F1 with semi-supervised training.

Conclusions

- Attention-based sequence transduction improves translation quality without recurrence or convolution and supports substantially more parallel training.
- Parsing results demonstrate transfer beyond translation, including training with approximately 40,000 annotated sentences.
- Full self-attention has quadratic sequence-length complexity; autoregressive generation remains sequential.
- Future directions include restricted attention for long sequences, non-text modalities, and less sequential generation.

References

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